# Experiment 6

Name: Renulasya
UID: **24BAI70210**
Branch: **CSE _AIML**
Section/Group: **24NTPP-2 A**
Sem: **5**
Subject: **Power BI**
Subject Code: **24CSH-341**

## OLAP Operations (Slice, Dice, Drill-up & Drill-down) using Microsoft Power BI

### Aim

To perform Dashboard and Scorecard Development using Microsoft Power BI, you need to create 5 graphs, 4 KPI cards, 3 slicers, and 1 business scorecard.

### Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. DAX (Data Analysis Expressions)

Dataset Source:
Global Superstore Dataset (Kaggle)
https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

### Theory

Online Analytical Processing (OLAP) is a Business Intelligence technique used for multidimensional analysis of business data. OLAP enables users to analyze information from different perspectives and perform complex analytical queries efficiently.

Common OLAP operations include:

Slice: Selecting a single dimension value from a dataset.
Dice: Selecting multiple dimensions and values simultaneously.
Drill-down: Moving from summarized data to detailed data.
Drill-up: Moving from detailed data to summarized data.

Power BI supports OLAP-style analysis through hierarchies, filters, slicers, visual interactions, and drill functionality.

### Conduct - Part A: Load Dataset and Build Report

Step 1: Open Power BI Desktop.
Step 2: Open the Power BI model created in Experiment 3 or import the Global Superstore dataset.
Step 3: Verify relationships between Orders, Returns, and People tables.

Step 4: Create report visuals:
• Sales by Region
• Profit by Category
• Sales by Segment
• Sales Trend by Year

## Conduct - Part B: Slice Operation

Step 5: Add a slicer visual using Region.
Step 6: Select a single Region such as West.
Step 7: Observe changes in Sales, Profit, and Orders.
Step 8: Record observations and capture screenshot evidence.

## Conduct - Part C: Dice Operation

Step 9: Apply multiple filters simultaneously.
Example:
• Region = West
• Category = Technology
• Year = 2022

Step 10: Analyze filtered business metrics.
Step 11: Record observations and capture screenshots.

## Conduct - Part D: Drill-down Operation

Step 12: Create a hierarchy using Region → State → City.
Step 13: Create a bar chart using the hierarchy.
Step 14: Use Drill-down mode to navigate from Region level to City level.
Step 15: Observe detailed sales patterns.
Step 16: Capture screenshots.

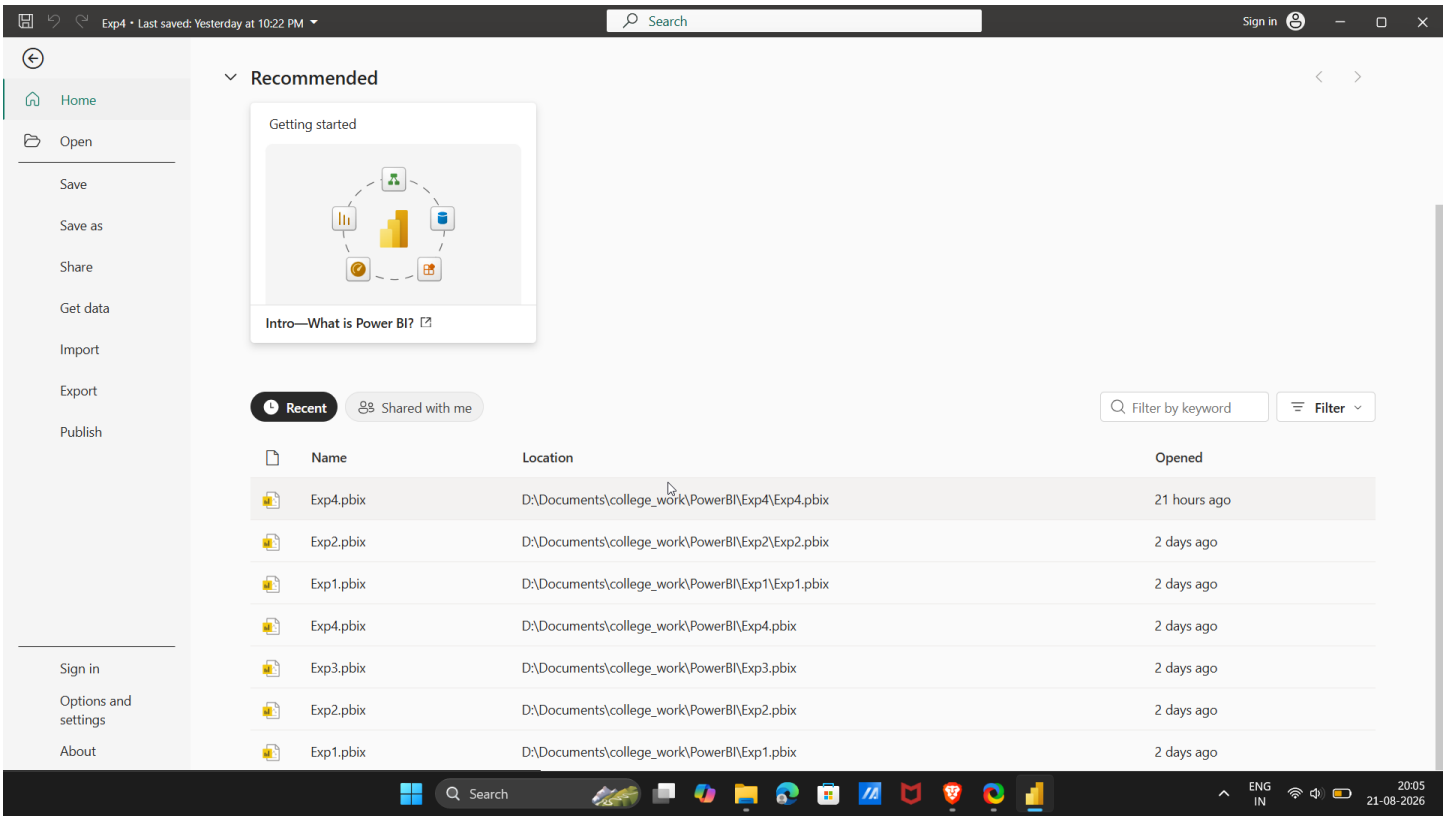
## Conduct - Part E: Drill-up Operation

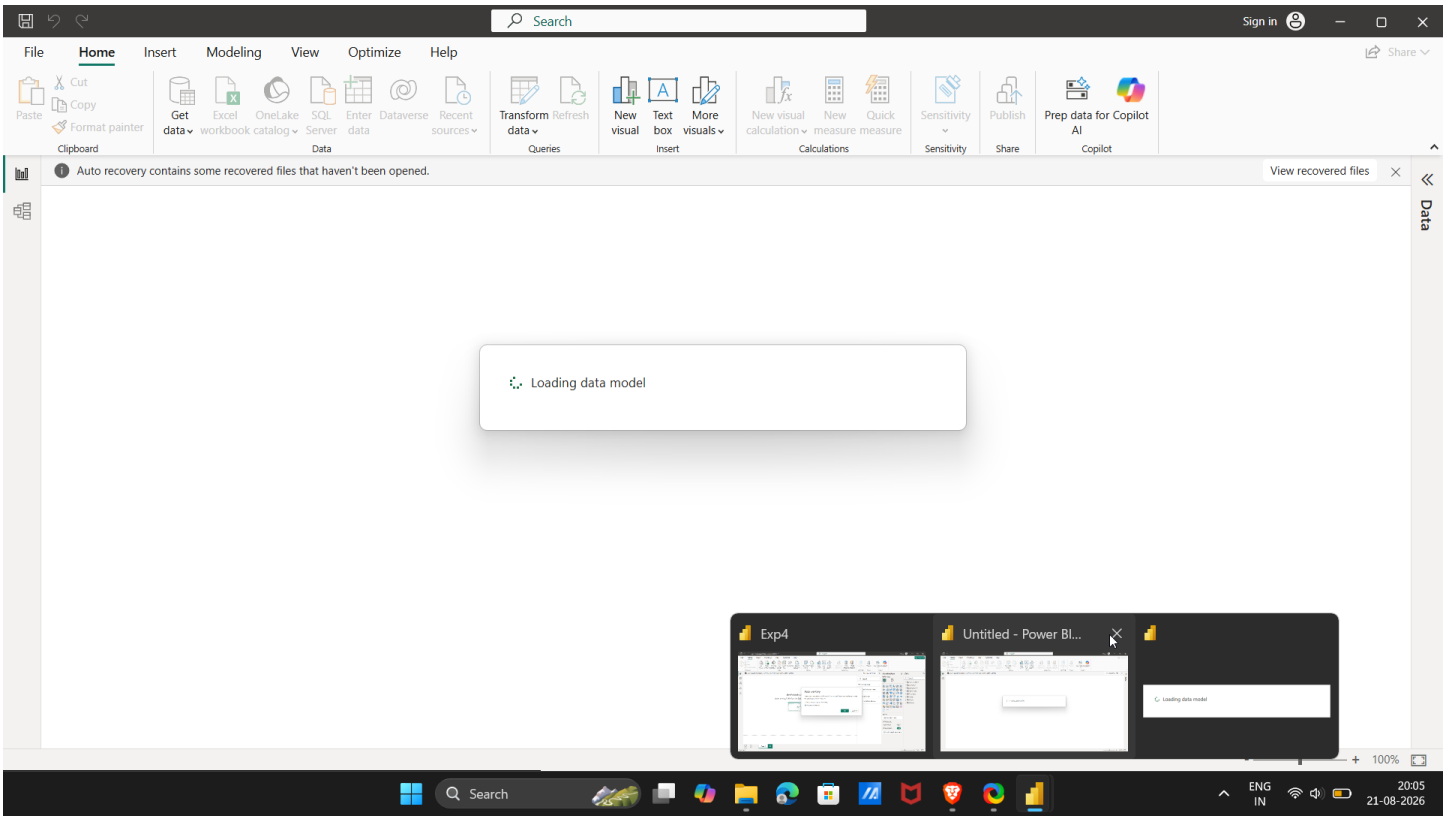
Step 17: Navigate back from City level to State level and then Region level.
Step 18: Compare aggregated values at each level.
Step 19: Record observations.
Step 20: Capture screenshots.

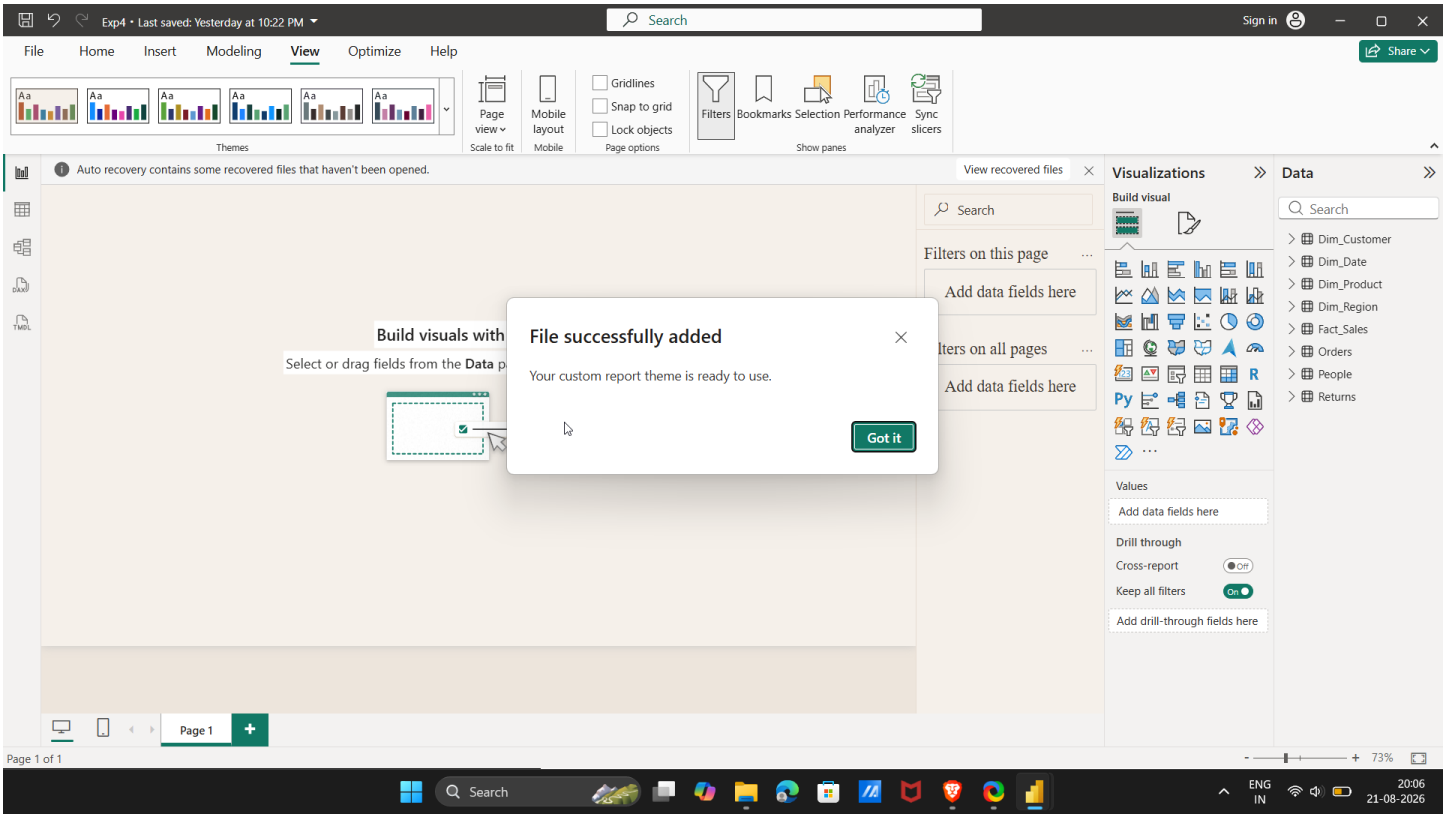
## Conduct - Part F: OLAP Analysis

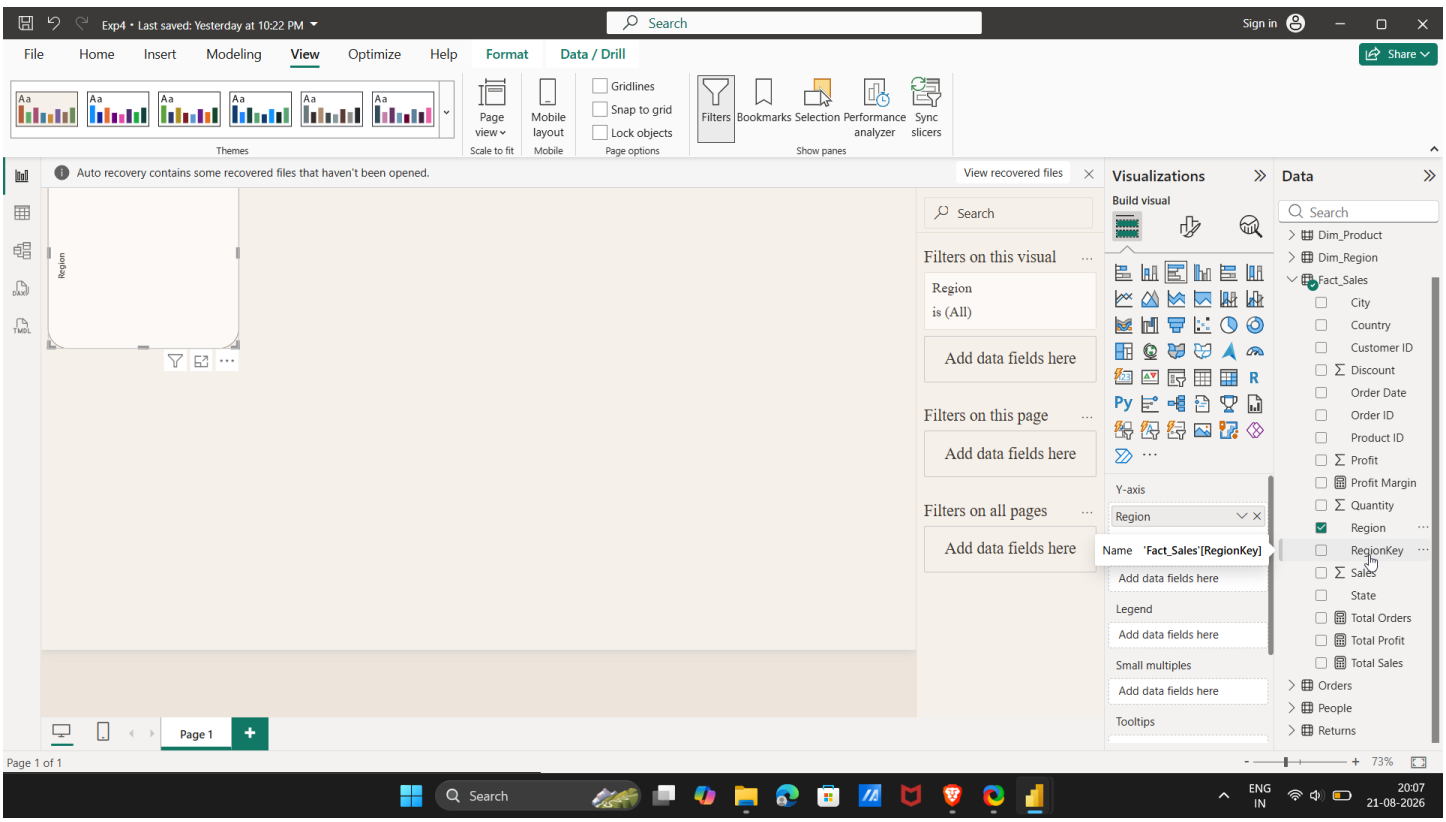
Step 21: Compare results obtained from Slice, Dice, Drill-down, and Drill-up operations.
Step 22: Create a summary table documenting observations.
Step 23: Explain how each operation supports business decision-making.

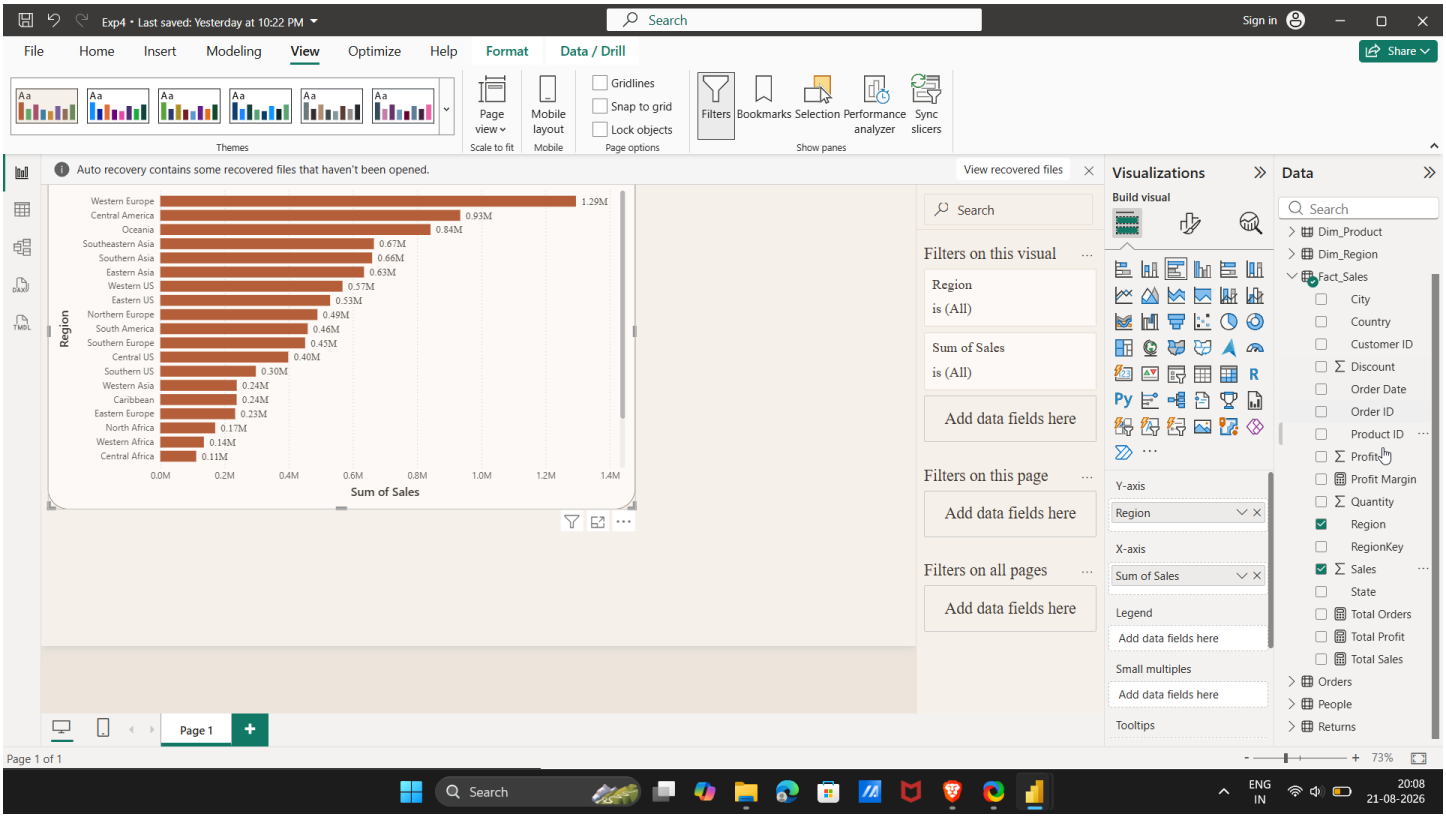
## Procedure

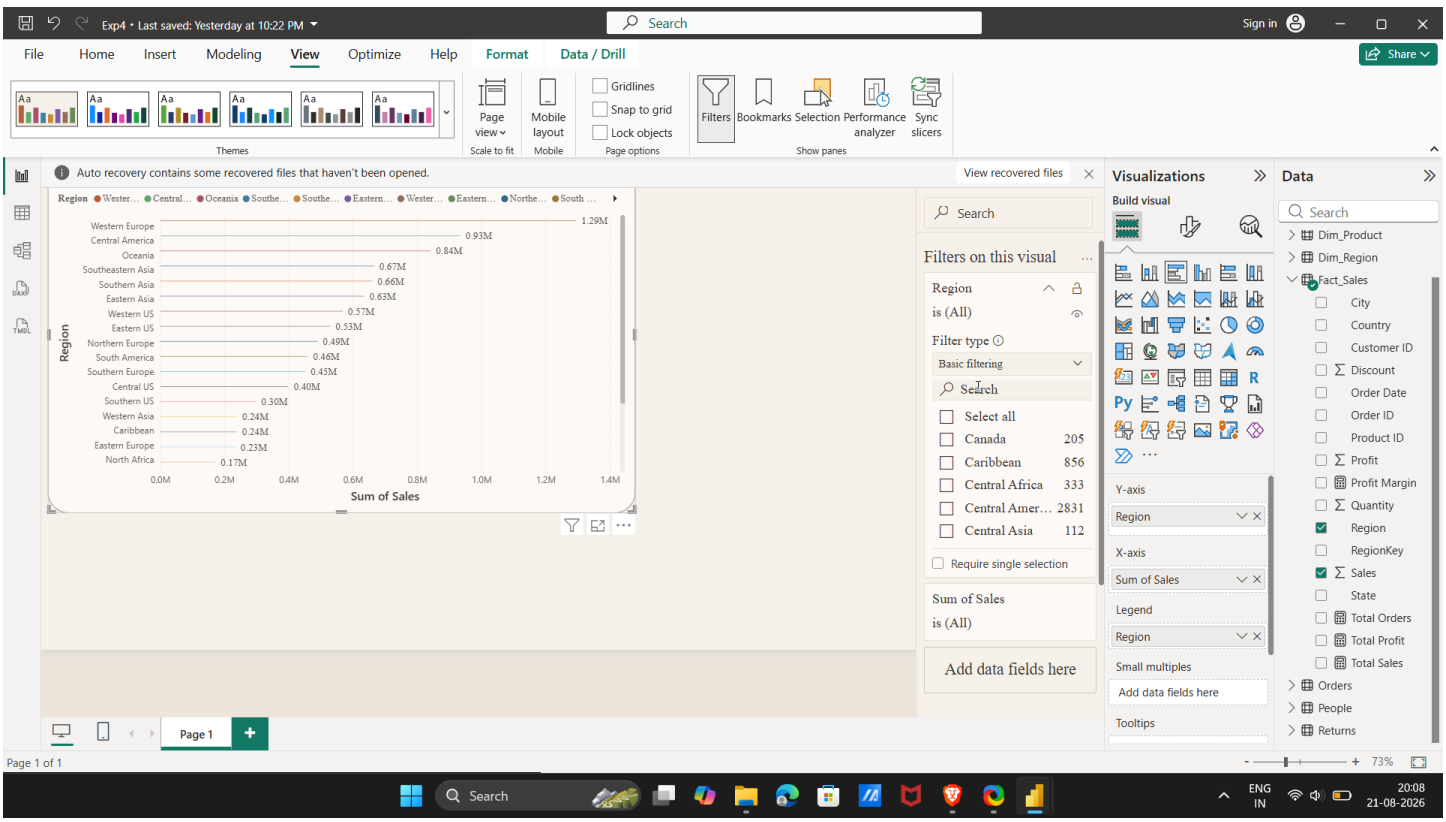

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
Home
Open
Save
Save as
Share
Get data
Import
Export
Publish
Sign in
Options and settings
About
Recommended
Getting started
Intro—What is Power BI?
Recent
Shared with me
Filter by keyword
Filter
Name
Location
Opened
Exp4.pbix D:\Documents\college_work\PowerBI\Exp4\Exp4.pbix 21 hours ago
Exp2.pbix D:\Documents\college_work\PowerBI\Exp2\Exp2.pbix 2 days ago
Exp1.pbix D:\Documents\college_work\PowerBI\Exp1\Exp1.pbix 2 days ago
Exp4.pbix D:\Documents\college_work\PowerBI\Exp4.pbix 2 days ago
Exp3.pbix D:\Documents\college_work\PowerBI\Exp3.pbix 2 days ago
Exp2.pbix D:\Documents\college_work\PowerBI\Exp2.pbix 2 days ago
Exp1.pbix D:\Documents\college_work\PowerBI\Exp1.pbix 2 days ago
ENG IN
20:05
21-08-2026

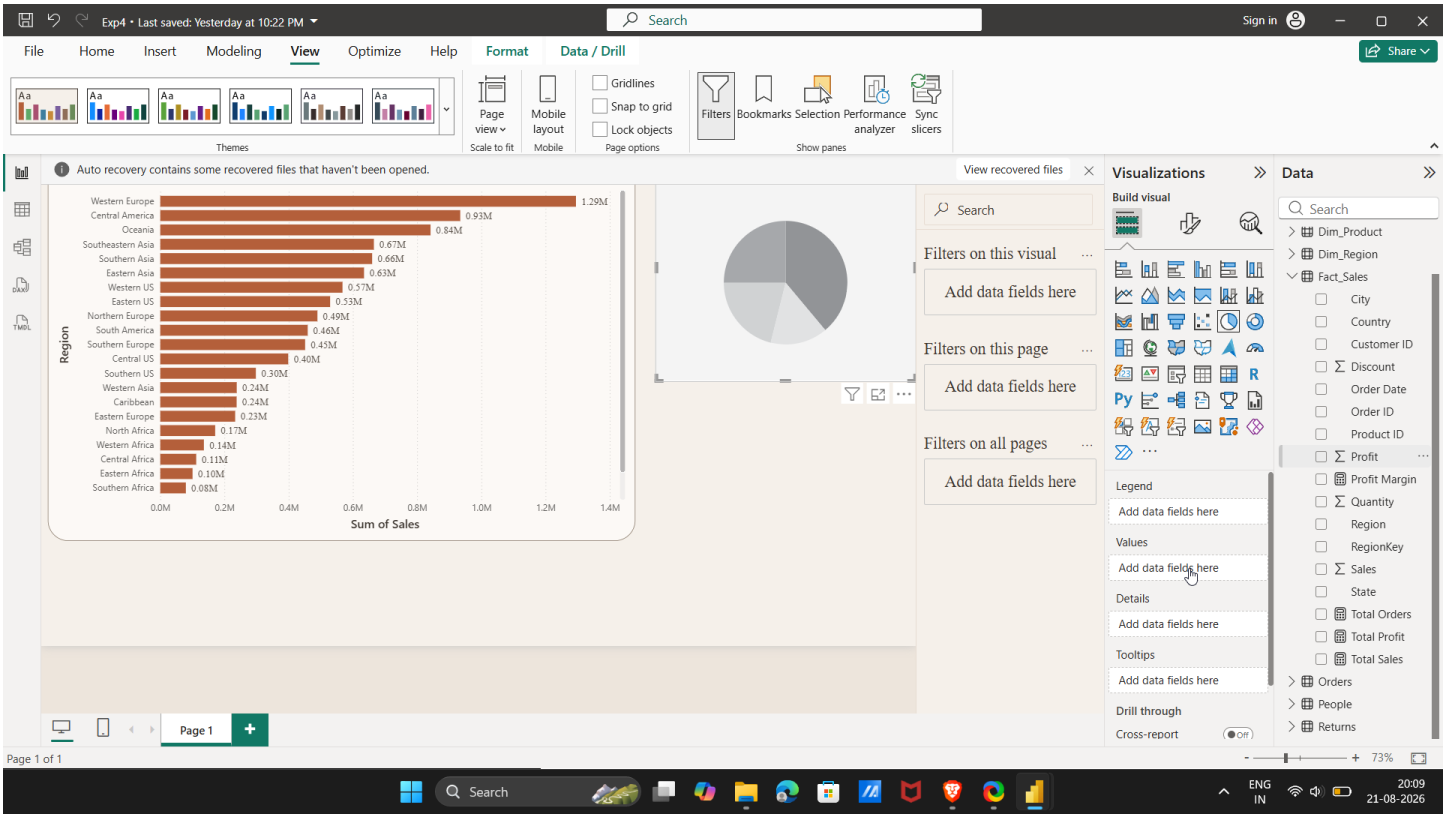

Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Prep data for Copilot AI
Copilot
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Data
Loading data model
Exp4
Untitled - Power BI...
100%
ENG IN
20:05
21-08-2026

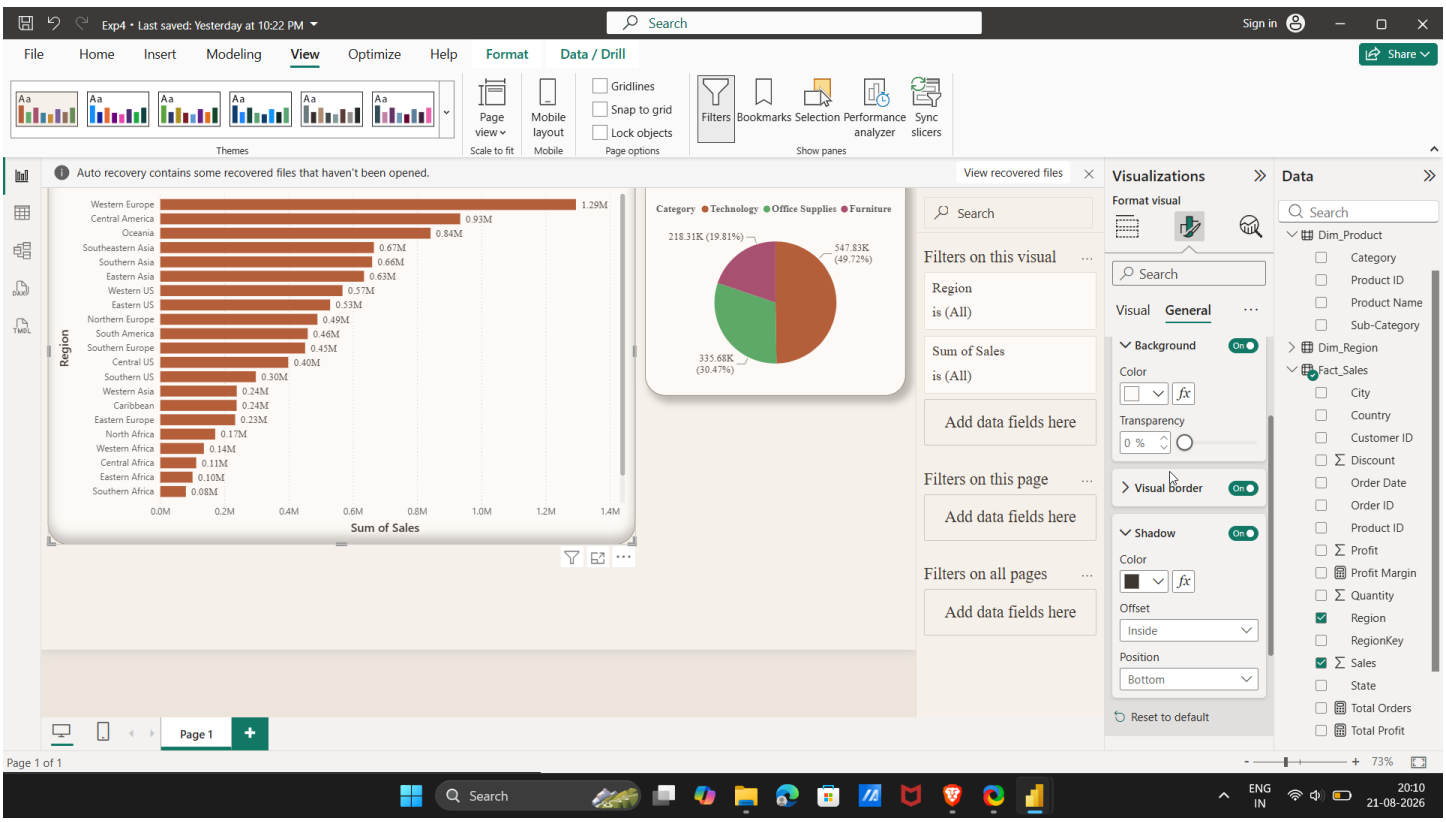

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Build visuals with
Select or drag fields from the Data p
File successfully added
Your custom report theme is ready to use.
Got it
Filters on this page
Add data fields here
Filters on all pages
Visualizations
Build visual
Values
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Dim_Customer
Dim_Date
Dim_Product
Dim_Region
Fact_Sales
Orders
People
Returns
Page 1
Page 1 of 1
73%
ENG
IN
20:06
21-08-2026

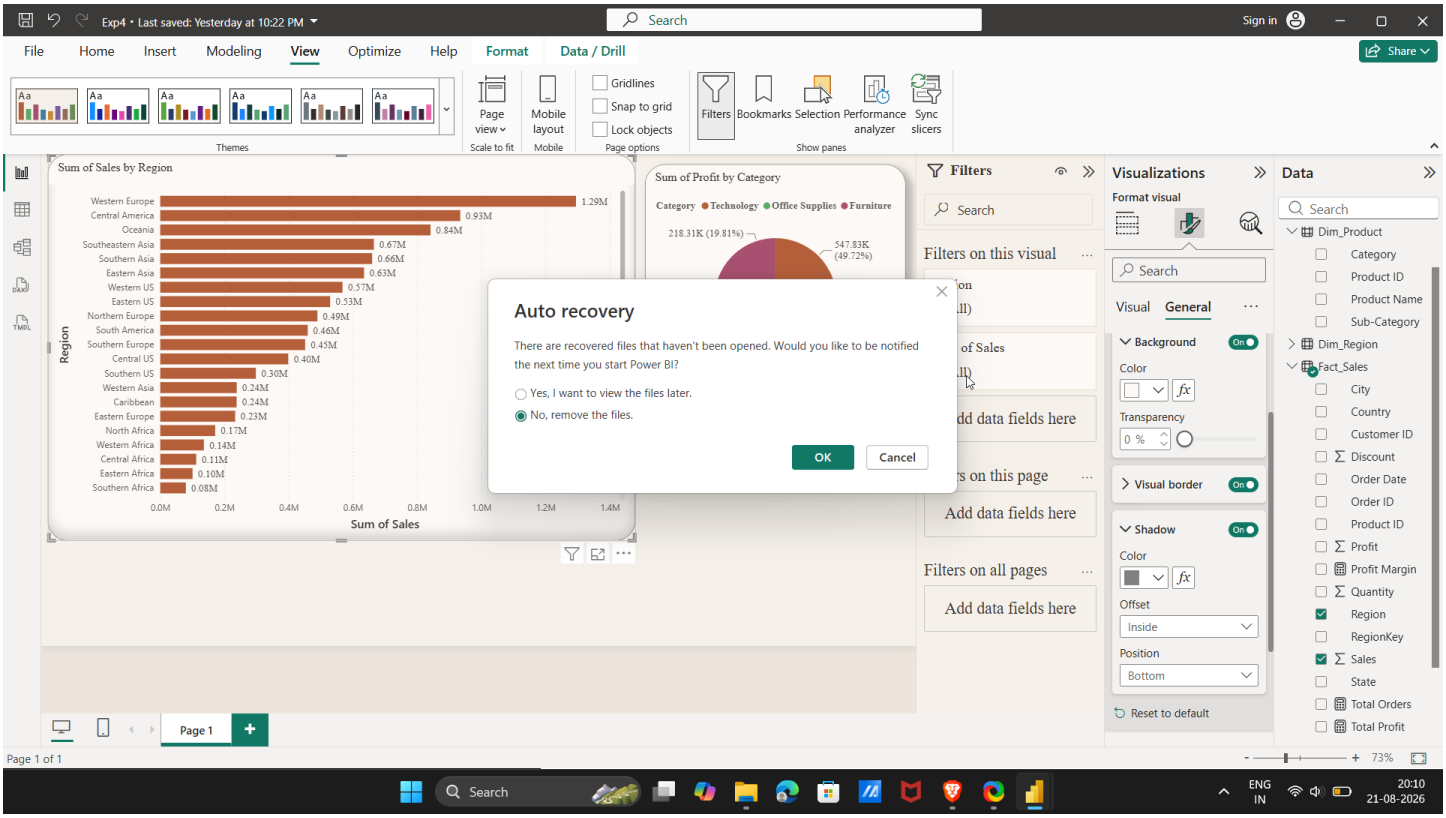

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Region
Filters on this visual
Region
is (All)
Add data fields here
Filters on this page
Filters on all pages
Visualizations
Build visual
Y-axis
Region
Name 'Fact_Sales'[RegionKey]
Legend
Small multiples
Tooltips
Data
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Total Sales
Orders
People
Returns
Page 1
Page 1 of 1
73%
ENG
IN
20:07
21-08-2026

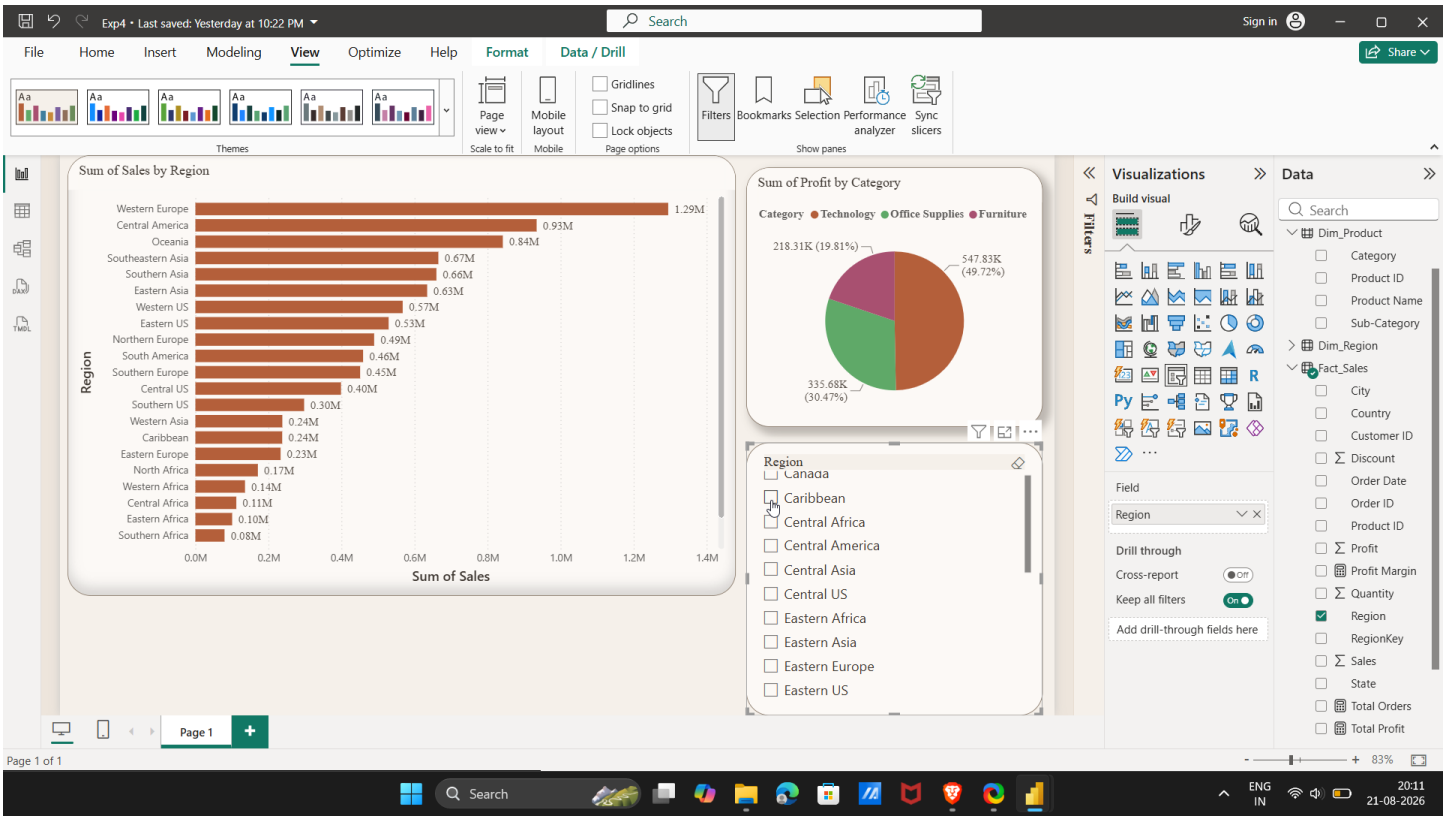

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Western Africa 0.14M
Central Africa 0.11M
Region
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Filters on this visual
Region
is (All)
Sum of Sales
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Y-axis
Region
X-axis
Sum of Sales
Legend
Add data fields here
Small multiples
Add data fields here
Tooltips
Data
Search
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Total Sales
Orders
People
Returns
Page 1
Page 1 of 1
73%
ENG IN
20:08
21-08-2026

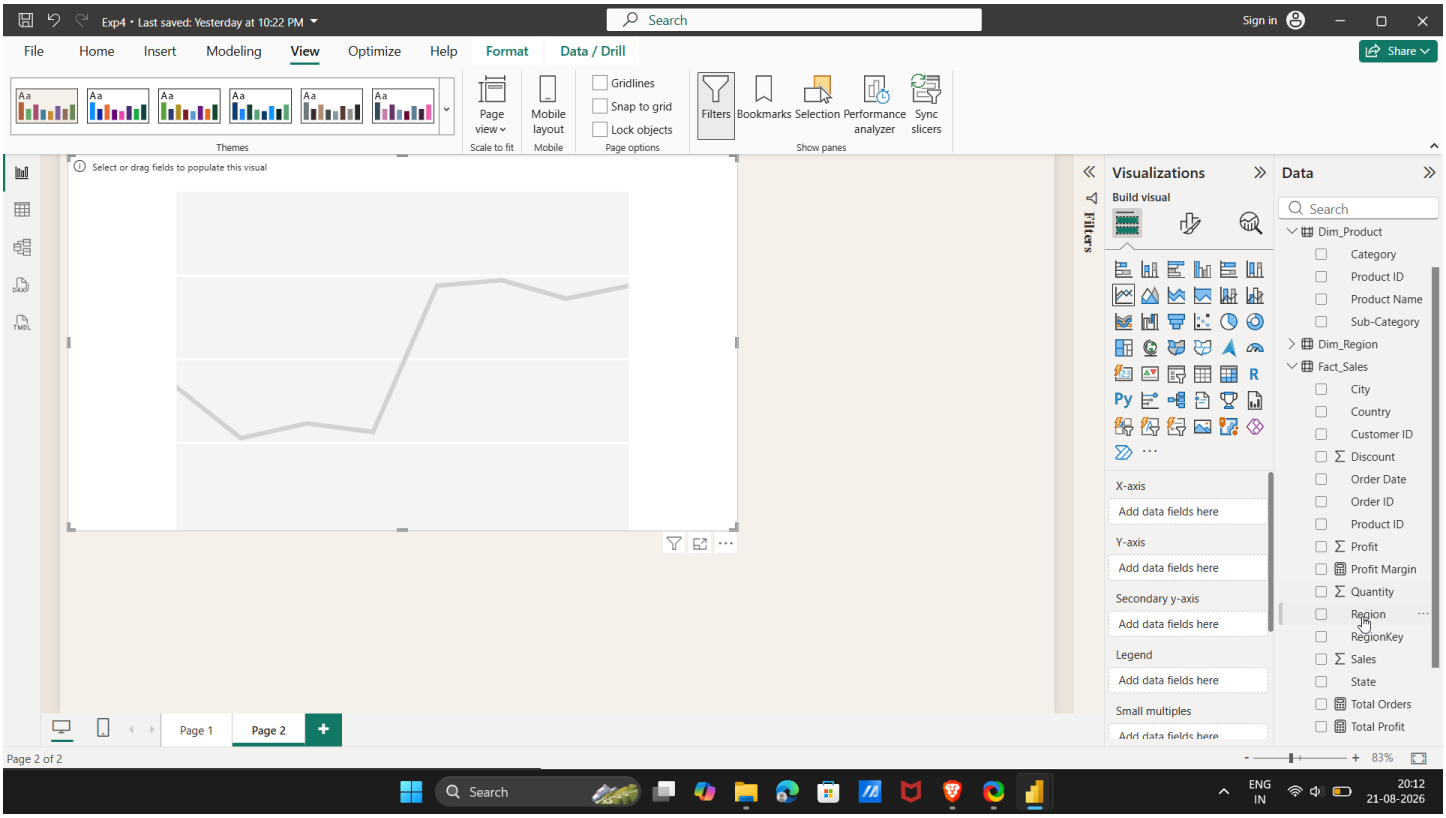

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Region Wester... Central... Oceania Southe... Southe... Eastern... Wester... Eastern... Northe... South ...
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Region
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Filters on this visual
Region
is (All)
Filter type
Basic filtering
Search
Select all
Canada 205
Caribbean 856
Central Africa 333
Central Amer... 2831
Central Asia 112
Require single selection
Sum of Sales
is (All)
Add data fields here
Visualizations
Build visual
Y-axis
Region
X-axis
Sum of Sales
Legend
Region
Small multiples
Add data fields here
Tooltips
Data
Search
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Total Sales
Orders
People
Returns
Page 1
Page 1 of 1
73%
ENG IN
20:08
21-08-2026

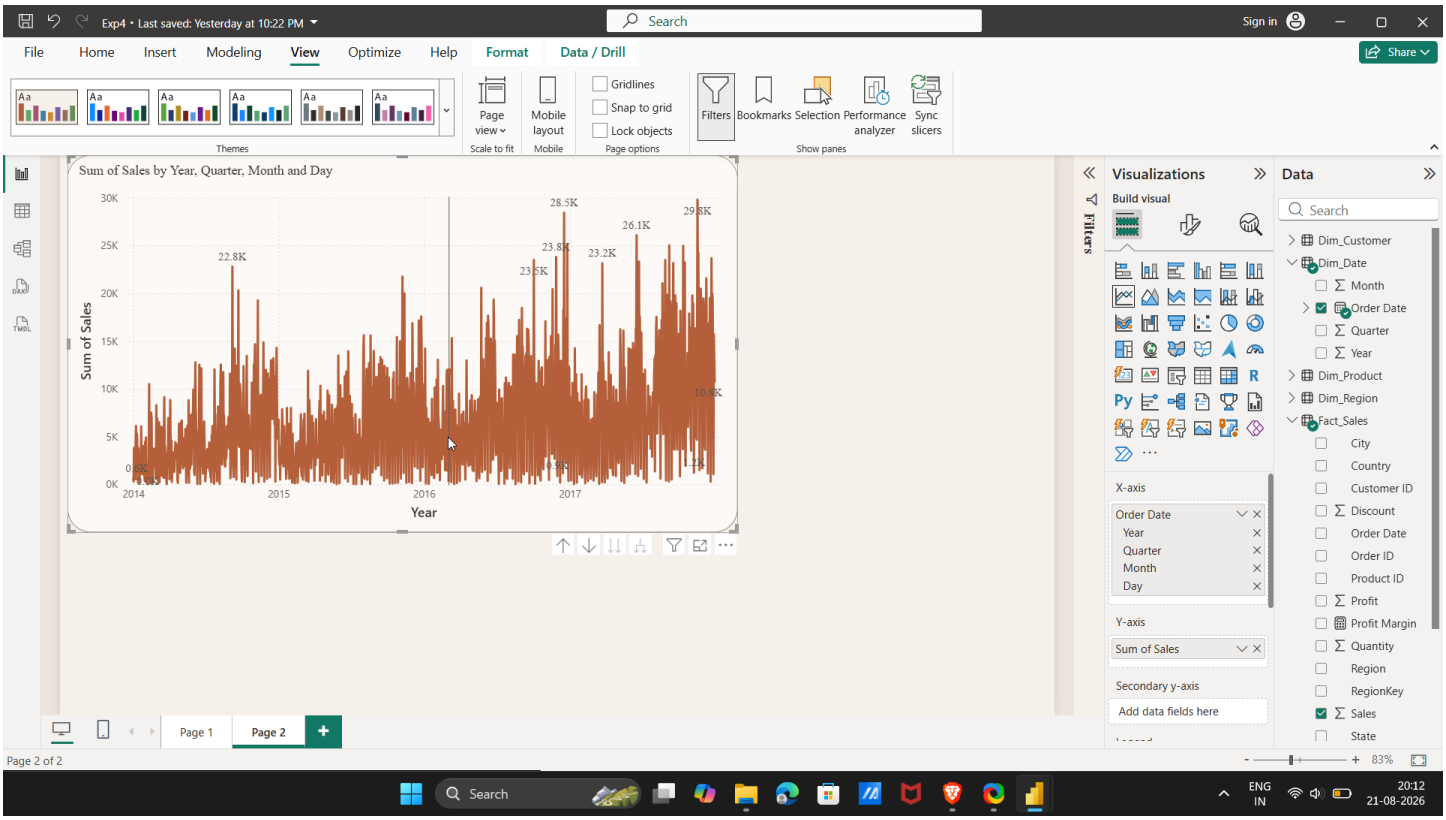

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Region
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Western Africa 0.14M
Central Africa 0.11M
Eastern Africa 0.10M
Southern Africa 0.08M
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Filters on this visual
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Legend
Add data fields here
Values
Add data fields here
Details
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Off
Data
Search
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Total Sales
Orders
People
Returns
Page 1
Page 1 of 1
73%
Search
ENG IN
20:09
21-08-2026

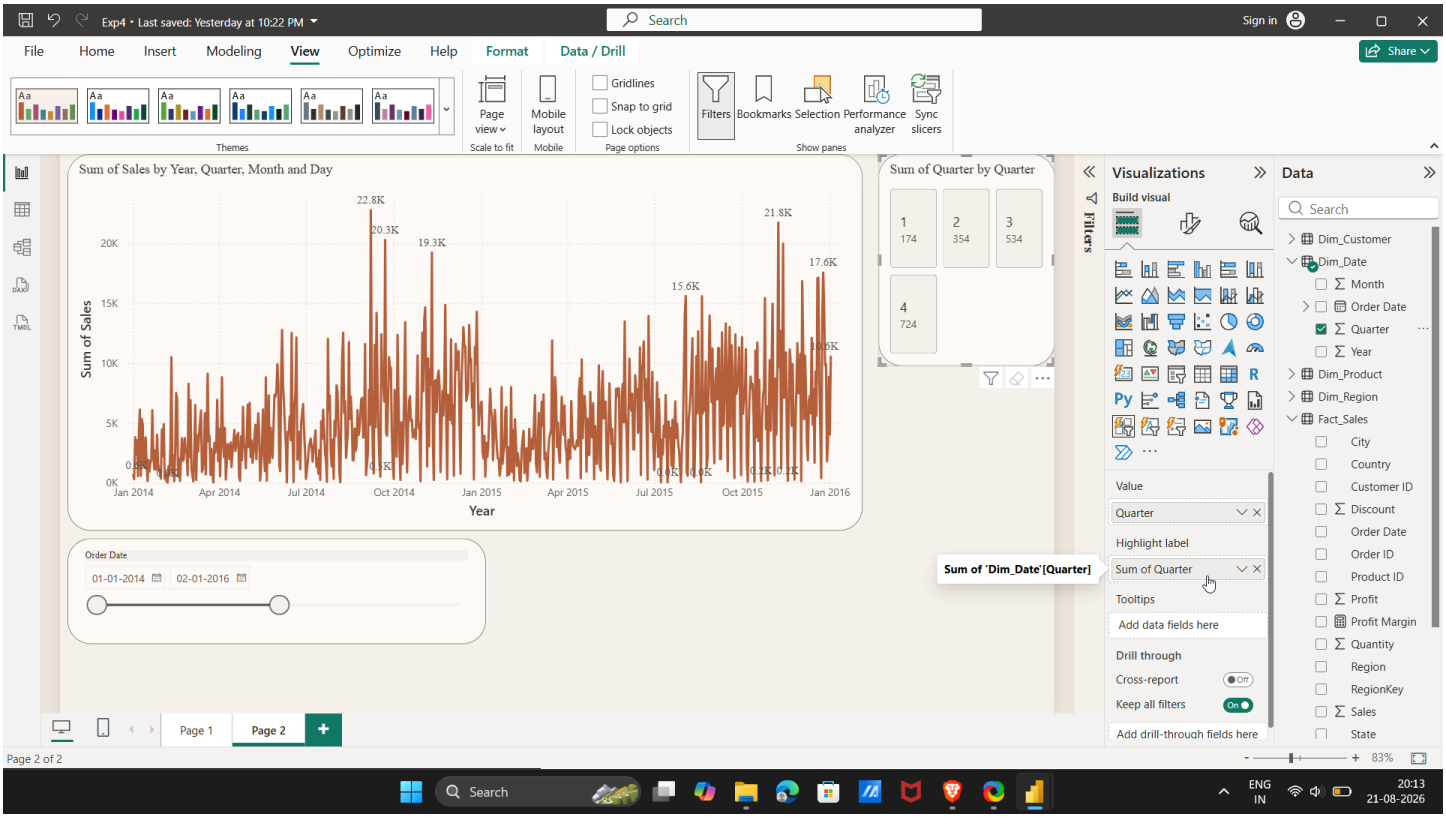

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Auto recovery contains some recovered files that haven't been opened.
View recovered files
Region
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Western Africa 0.14M
Central Africa 0.11M
Eastern Africa 0.10M
Southern Africa 0.08M
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Category Technology Office Supplies Furniture
218.31K (19.81%)
547.83K (49.72%)
335.68K (30.47%)
Filters on this visual
Region is (All)
Sum of Sales is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Format visual
Search
Visual General
Background On
Color
Transparency
0 %
Visual Border On
Shadow On
Color
Offset
Inside
Position
Bottom
Reset to default
Data
Search
Dim_Product
Category
Product ID
Product Name
Sub-Category
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Page 1
Page 1 of 1
73%
Search
ENG IN
20:10
21-08-2026

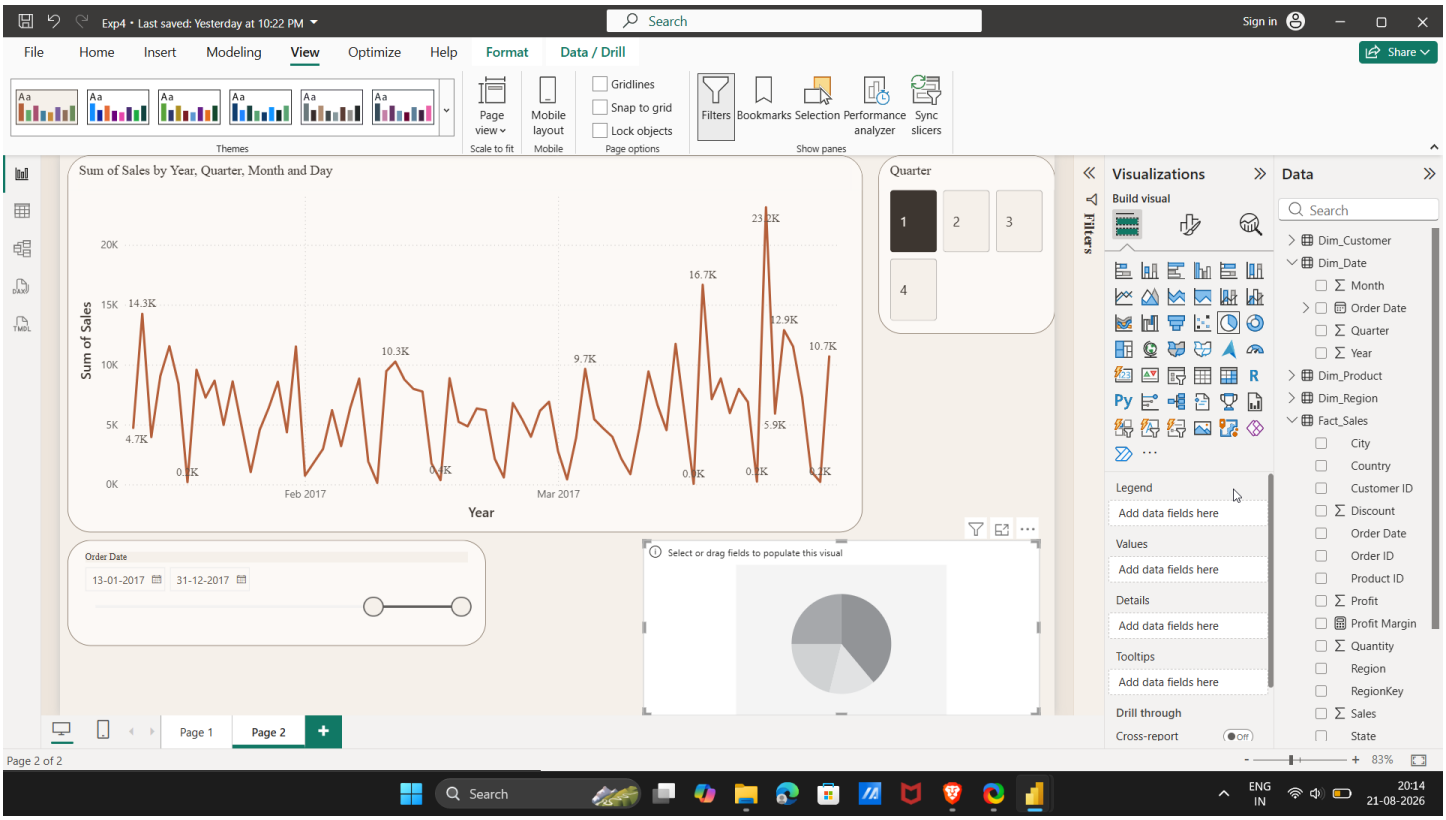

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Themes
Scale to fit
Mobile
Page options
Show panes
Sum of Sales by Region
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Western Africa 0.14M
Central Africa 0.11M
Eastern Africa 0.10M
Southern Africa 0.08M
Region
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Sum of Profit by Category
Category Technology Office Supplies Furniture
218.31K (19.81%)
547.83K (49.72%)
Auto recovery
There are recovered files that haven't been opened. Would you like to be notified the next time you start Power BI?
Yes, I want to view the files later.
No, remove the files.
OK
Cancel
Filters
Search
Filters on this visual
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Format visual
Visual General
Background On
Color
Transparency
0 %
Visual border On
Shadow On
Color
Offset
Inside
Position
Bottom
Reset to default
Data
Search
Dim_Product
Category
Product ID
Product Name
Sub-Category
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Page 1
Page 1 of 1
73%
Search
ENG IN
20:10
21-08-2026

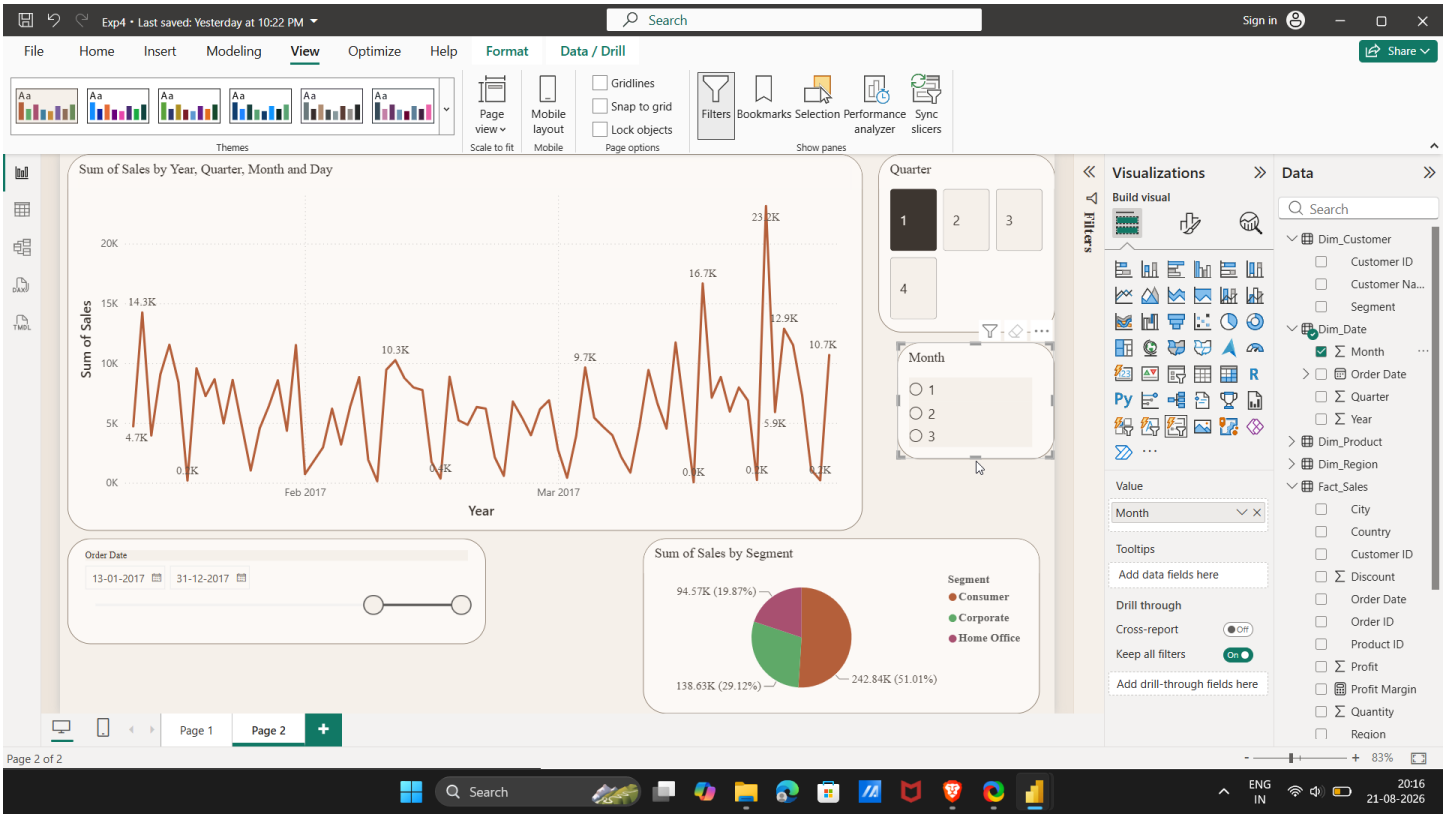

Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Themes
Scale to fit
Mobile
Page options
Show panes
Sum of Sales by Region
Western Europe 1.29M
Central America 0.93M
Oceania 0.84M
Southeastern Asia 0.67M
Southern Asia 0.66M
Eastern Asia 0.63M
Western US 0.57M
Eastern US 0.53M
Northern Europe 0.49M
South America 0.46M
Southern Europe 0.45M
Central US 0.40M
Southern US 0.30M
Western Asia 0.24M
Caribbean 0.24M
Eastern Europe 0.23M
North Africa 0.17M
Western Africa 0.14M
Central Africa 0.11M
Eastern Africa 0.10M
Southern Africa 0.08M
Region
0.0M 0.2M 0.4M 0.6M 0.8M 1.0M 1.2M 1.4M
Sum of Sales
Sum of Profit by Category
Category Technology Office Supplies Furniture
218.31K (19.81%)
547.83K (49.72%)
335.68K (30.47%)
Region
Canada
Caribbean
Central Africa
Central America
Central Asia
Central US
Eastern Africa
Eastern Asia
Eastern Europe
Eastern US
Filters
Visualizations
Build visual
Field
Region
Drill through
Cross-report Off
Keep all filters On
Add drill-through fields here
Data
Search
Dim_Product
Category
Product ID
Product Name
Sub-Category
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
Page 1
Page 1 of 1
83%
Search
ENG IN
20:11
21-08-2026


Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Select or drag fields to populate this visual
Visualizations
Build visual
Data
Search
Dim_Product
Category
Product ID
Product Name
Sub-Category
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Total Orders
Total Profit
X-axis
Add data fields here
Y-axis
Add data fields here
Secondary y-axis
Add data fields here
Legend
Add data fields here
Small multiples
Add data fields here
Page 1
Page 2
Page 2 of 2
83%
Search
ENG IN
20:12
21-08-2026


Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Themes
Page view
Mobile layout
Gridlines
Snap to grid
Lock objects
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Scale to fit
Mobile
Page options
Show panes
Sum of Sales by Year, Quarter, Month and Day
Sum of Sales
Year
30K
25K
20K
15K
10K
5K
0K
2014
2015
2016
2017
22.8K
28.5K
23.8K
23.5K
23.2K
26.1K
29.8K
10.8K
Visualizations
Build visual
Data
Search
Dim_Customer
Dim_Date
Month
Order Date
Quarter
Year
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
X-axis
Order Date
Year
Quarter
Month
Day
Y-axis
Sum of Sales
Secondary y-axis
Add data fields here
Page 1
Page 2
Page 2 of 2
83%
Search
ENG IN
20:12
21-08-2026


Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Sum of Sales by Year, Quarter, Month and Day
Sum of Sales
Year
Sum of Quarter by Quarter
Order Date
01-01-2014
02-01-2016
Visualizations
Build visual
Value
Quarter
Highlight label
Sum of Quarter
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Sum of 'Dim_Date'[Quarter]
Data
Search
Dim_Customer
Dim_Date
Month
Order Date
Quarter
Year
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Page 1
Page 2
Page 2 of 2
83%
ENG IN
20:13
21-08-2026


Exp4 • Last saved: Yesterday at 10:22 PM
Search
Sign in
File Home Insert Modeling View Optimize Help Format Data / Drill
Share
Sum of Sales by Year, Quarter, Month and Day
Sum of Sales
Year
Quarter
Order Date
13-01-2017
31-12-2017
Select or drag fields to populate this visual
Visualizations
Build visual
Legend
Add data fields here
Values
Add data fields here
Details
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Data
Search
Dim_Customer
Dim_Date
Month
Order Date
Quarter
Year
Dim_Product
Dim_Region
Fact_Sales
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin
Quantity
Region
RegionKey
Sales
State
Page 1
Page 2
Page 2 of 2
83%
ENG IN
20:14
21-08-2026




## Output

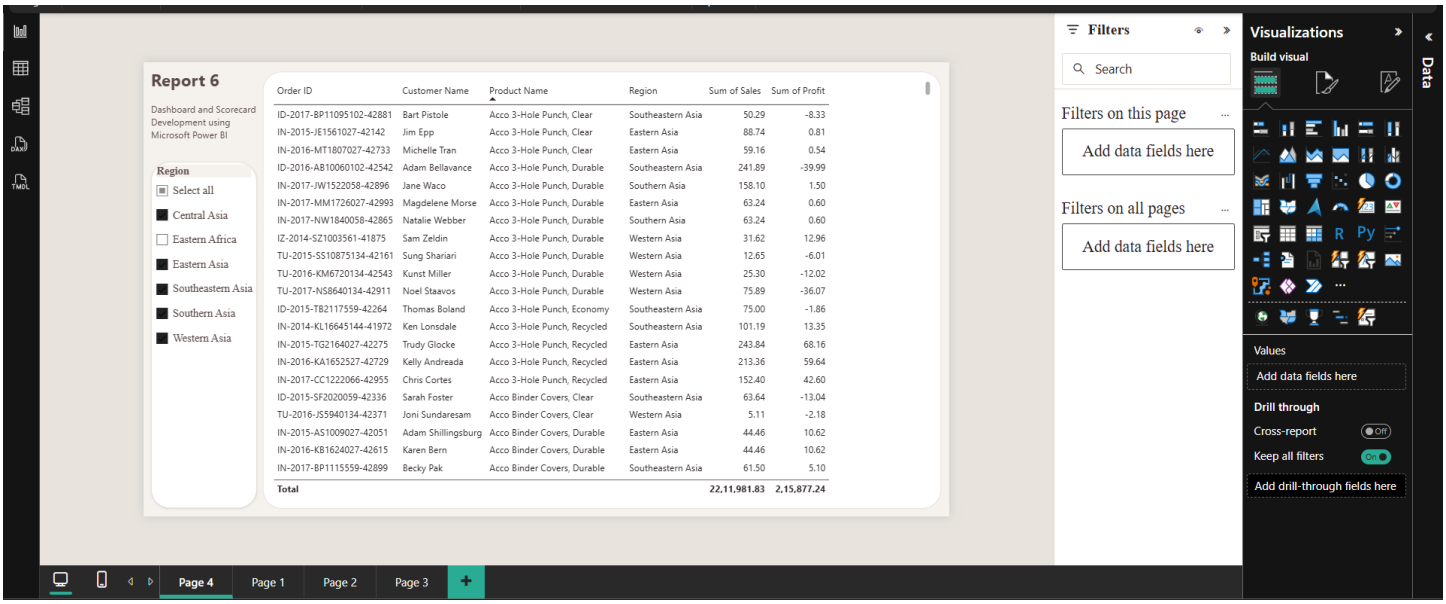
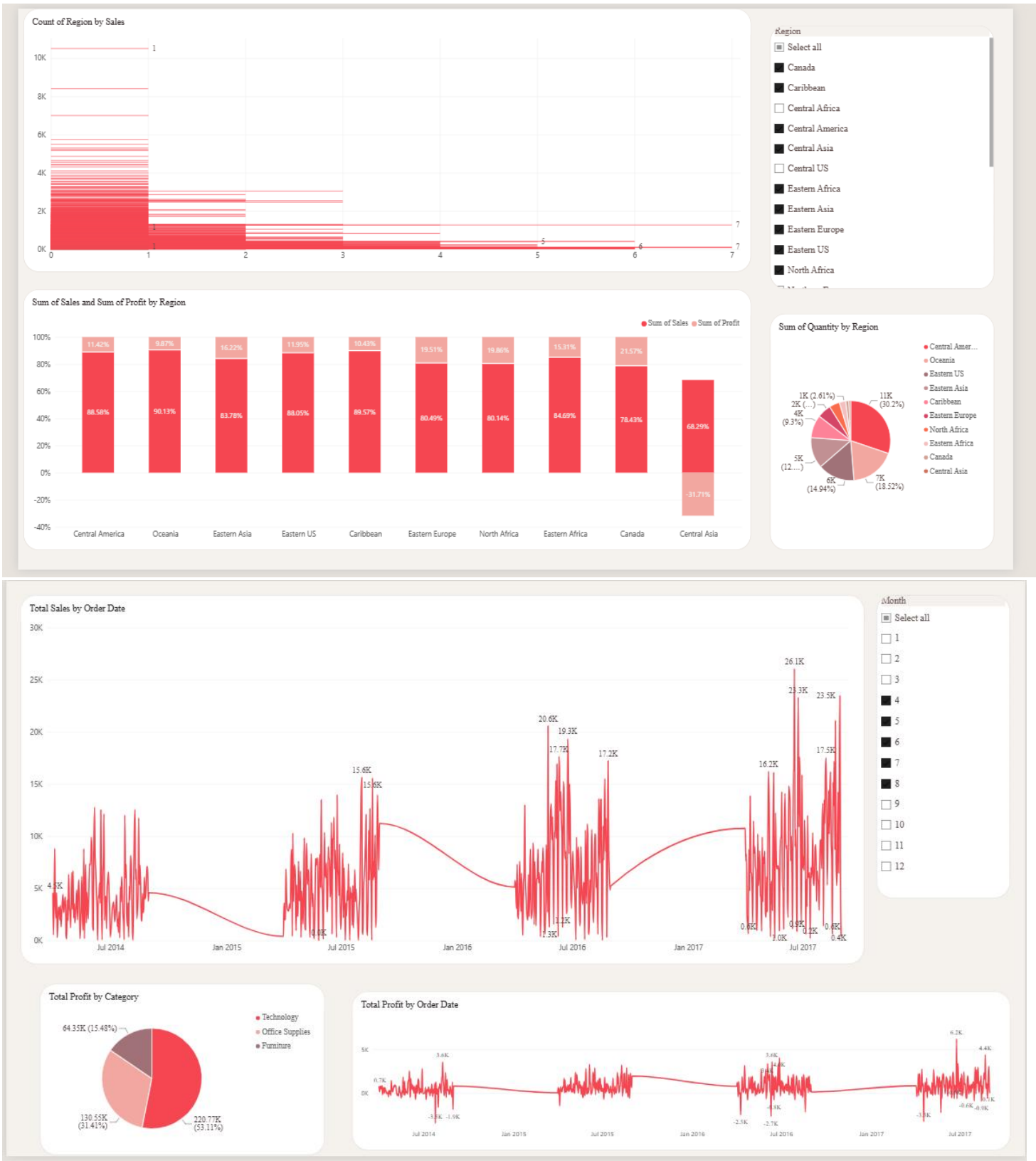



**Figure A:** Cluster bar chart and Donut chart


Count of Region by Sales
Region
Select all
Canada
Caribbean
Central Africa
Central America
Central Asia
Central US
Eastern Africa
Eastern Asia
Eastern Europe
Eastern US
North Africa
Sum of Sales and Sum of Profit by Region
Sum of Sales
Sum of Profit
Central America
Oceania
Eastern Asia
Eastern US
Caribbean
Eastern Europe
North Africa
Eastern Africa
Canada
Central Asia
Sum of Quantity by Region
Total Sales by Order Date
Month
Select all
Total Profit by Category
Technology
Office Supplies
Furniture
Total Profit by Order Date

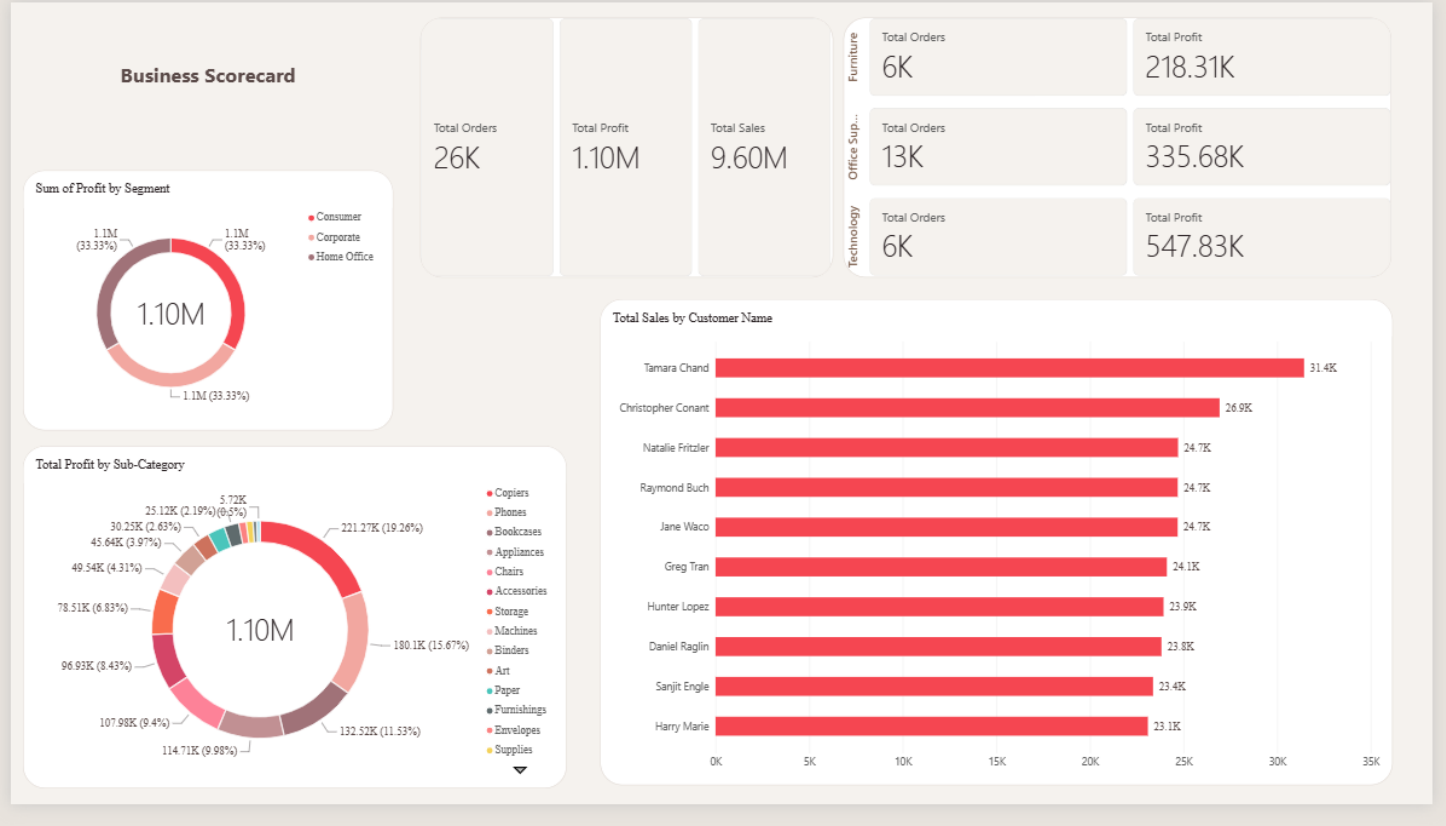



**Figure B:** Line Chart with Slicer

## Result

OLAP operations including Slice, Dice, Drill-down, and Drill-up were successfully performed using Microsoft Power BI. Multidimensional analysis was carried out on the Global Superstore dataset, enabling detailed and summarized views of business performance from multiple perspectives.

## Learning Outcomes

- Explain OLAP concepts and multidimensional analysis. (CO3-BT3)
- Perform Slice and Dice operations using filters and slicers. (CO3-BT3)
- Create hierarchies and perform Drill-down and Drill-up analysis. (CO5-BT6)
- Interpret analytical insights generated through OLAP operations. (CO3-BT4)
- Design interactive analytical reports for business users. (CO5-BT6)

## Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics.
2. Ralph Kimball & Margy Ross, The Data Warehouse Toolkit.
3. Cindi Howson, Successful Business Intelligence.
4. Microsoft Power BI Documentation.